9. Domain: Power Electronics and Renewable Energy

Sr. No.	Parameters	Brief details
1.	**Brief details of Technology Developed**	Development of three-phase voltage source inverter for Variable Frequency Drive of Permanent Magnet-less Synchronous Reluctance Motors for speed control applications
	Name of the Project under which technology is developed	Exploratory Research under C-SPARK (2023-24 cycle)
	Funding Agency	Internal Funding
	Technology Readiness Level (TRL)	TRL-4
	Brief Write-up on the Technology Developed	This project concerns on the design and implementation of three-phase voltage source inverter (VSI) for variable frequency drive. The developed drive would be used to generate variable frequency output suitable to be fed to the Synchronous Reluctance Motors (SyRM) for the purpose of variable speed control which finds applications in process industries and electric vehicles. The SyRM has niether permanent magnets (or) windings in its rotor. Compared to the interior permanent magnet Synchronous motor (IPMSM), SyRM maintains most of the desirable features without using the permanent magnet and thus achieves lower cost and better overload handling ability. The presence of large rotor saliency makes the SyRM very suitable for high-speed drives that require field-weakening (FW) and wide constant power-speed ratio (CPSR) for drives in electric vehicles and traction applications. The developed VSI comprises uncontrolled diode bridge rectifier at it front-end for AC-DC rectification with dc link precharging, dc link capacitors to smooth out pulsating DC voltage from a rectifier, stabilizing the DC bus, absorbing voltage spikes, and providing ride-through capability during grid voltage sags. The inverter equipped with four leg IGBT based DC-AC conversion. The fourth leg is used as brake chopper to prevent uncontrolled increase of DC link voltage. The VSI stack consists of Diode modules, dc link capacitors with DC link measurement terminals, brake chopper module with external brake resistor connectivity, IGBT-based inverter with gate driver PCBs, Heat Sink, dual inlet Fan for forced air cooling, RC Snubber, Bus bar and Thermal trip, heavy-duty terminal connectors. The drive has the provision to connect the external DC source directly, bypassing diode rectifier, to supply the inverter.
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